SDC

Hackathon · Technical

Technical Tools Declaration

Lumina Nusantara · Domain 1 — The Aiversecity Future-Ready Workforce Domain

SDG Track: SDG 4 — Quality Education (sub-target 4.4, with 8.5)

Team Leader: DU YULIN · Xiamen University Malaysia (XMUM)

Live prototype: https://sdc2026-lumina-nusantara-65cnugvuxdle5dgp9ky3fs.streamlit.app

Code repository: https://github.com/douyinlin/sdc2026-lumina-nusantara

Submission deadline: 25 October 2026

> Technical Tools Declaration. Complete every section; where a category does not apply we ticked "No" or wrote "None" rather than leaving it blank.

1. TEAM AND PROJECT OVERVIEW

Team Name	Lumina Nusantara
Project / Solution Name	SkillRadar MY
Name (Leader)	DU YULIN
Email & Phone (Leader)	CYS2609009@xmu.edu.my / +60 11-1798 1756
University Affiliation(s)	Xiamen University Malaysia (XMUM)
Live Prototype URL	https://sdc2026-lumina-nusantara-65cnugvuxdle5dgp9ky3fs.streamlit.app
Code Repository URL (public)	https://github.com/douyinlin/sdc2026-lumina-nusantara

2. ARCHITECTURE & TECHNOLOGY STACK

2.1 System Architecture Overview

The Streamlit user interface is the only client judges and users touch. A user either uploads a synthetic CV or edits the questionnaire, both of which produce a skill profile normalised against a fixed vocabulary (feature #1). The UI reads a versioned trend snapshot from the data store; each record carries its source URL and retrieval date, and these are displayed alongside every forecast (feature #2). The matching and pathway engine consumes the profile plus the snapshot and returns ranked roles with a four-component explainable score, the exact skill gap (feature #3), an ordered reskilling plan (feature #4), SDG tags on every role (feature #5) and a moderate-versus-fast scenario comparison (feature #6). CV parsing and scoring run locally in the app with no external AI service, with a local rule-based fallback so the demo degrades gracefully. A scheduled ingestion job refreshes trend snapshots from public sources. See the appended block diagram.

2.2 Languages, Frameworks & Libraries

Name & Version	Purpose in Project	License
Python 3.11+	Application and engine language	PSF
Streamlit 1.66	Web UI, file upload, tables	Apache 2.0
pandas 2.x	Tabular handling	BSD 3-Clause
reportlab	Build these submission PDFs (offline tool)	BSD
None (no ML library loaded at runtime)	All scoring is a deterministic rule set	Apache 2.0
unittest (stdlib)	Unit tests for the scoring engine	PSF

3. APIs & CLOUD SERVICES

3.1 External APIs

API and Provider	Purpose in Project	Access Tier
None — no external AI API is called	CV parsing and scoring run locally in the app	—
WEF / O*NET / HRD Corp public data feeds	Trend snapshot ingestion (public data)	free
Streamlit Community Cloud	Hosting the public live demo	free
—	—	—

No API keys or secrets are included in this document.

3.2 Cloud Platforms & Deployment

Service & Provider	Purpose in Project	Plan / Region
Streamlit Community Cloud	Public live demo (auto-deploy from GitHub main)	Community / global
Replit	Development workspace and alternative hosting	Free tier
Streamlit Community Cloud	Auto-deploy from GitHub main on every push	Community / global
GitHub (douyinlin)	Source control and deployment trigger	Free

4. AI TOOLS, MODELS & DATASETS

4.1 AI Models Used in the Solution

Model & Version	Source / Provider	Usage	License / Terms
No pre-trained model is loaded at runtime	—	—	—
Rule-based skill extractor (in-house)	Team-built	rules only	Apache 2.0
Dictionary-matched taxonomy (O*NET / ESCO inspired)	Team-curated	catalogue	CC BY 4.0 (O*NET)

4.2 Datasets

Dataset	Source / URL	Type	License	Personal data?
O*NET skills & occupations	onetcenter.org	public	CC BY 4.0	No
WEF Future of Jobs indicators	weforum.org	public	Cited	No
HRD Corp course directory	hrdcorp.gov.my	public	Public directory	No
Demo profiles & sample CV	Team-authored	synthetic	Apache 2.0	No

4.3 AI-Assisted Development Tools

Did your team use any generative-AI tool to help create code, design, text, or other project material? **Yes**.

Tool & Version	Used For	Extent of Use
DeepSeek-based coding assistant	Scaffolding, boilerplate, documentation drafting	moderate
GitHub Copilot	Inline code completion	minor

Review statement: Every AI-generated block was read, tested or rewritten by a team member; matching logic, SDG tags and all statistics are hand-verified against the cited sources. Unit tests cover the scoring and pathway functions, and a reviewer other than the author checks each pull request before merge.

5. SPONSOR-PROVIDED RESOURCES

Did your project use any official SDC sponsor-provided resource? **Yes**.

Sponsor	Resource	How It Was Used in the Project
SDC 2026 programme	Replit workspace	Development environment for the prototype during the build phase
None	No RunPod endpoint is used	CV parsing runs locally; the app makes no GPU or external AI call

Essentiality: The Replit workspace removed local setup so all five members could work on the same running prototype, and the public demo is hosted on Streamlit Community Cloud. Neither is essential to the logic: the app runs on any Python host with no external AI service, so the solution survives if workspace access ends.

6. SECURITY, PRIVACY & RESPONSIBLE USE

6.1 Data Handling & Privacy

The public demo processes no real personal data. Users either pick a synthetic sample profile or upload a synthetic CV that the team authored; the interface states this rule next to the upload widget. Uploaded text is parsed in memory to extract skill names and is never written to disk, logged or stored in a database. No account, login or cookie is used beyond the Streamlit session itself. API keys live in platform secrets (Replit Secrets / Streamlit Cloud secrets) and never appear in the repository. If a future version accepts real CVs, it will require explicit consent, a retention limit and deletion on request.

6.2 Known Limitations & Risks

Main risks: (1) trend data lag and source bias — the weights are currently simulated and the UI labels them as such; (2) rule-based matching can mis-map niche skills — mitigated by a curated alias vocabulary and a visible skill-gap breakdown; (3) hallucinated roles or courses — impossible by construction, because roles and course links come from a fixed catalogue and no language model runs at inference time; (4) no GPU acceleration today, so CV parsing is limited to text-based PDFs and plain text; (5) small pilot size — statements are labelled as estimates. The tool supports human career decisions and never makes hiring decisions.

7. COMPLIANCE & READINESS

- ■ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ■ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ■ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ■ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ■ Any use of official SDC sponsor resources has been declared in Section 5.
- ■ The prototype described here is accessible to judges through the links provided in Section 1.
- ■ All fields in this template have been completed within the word limits stated.

8. DECLARATION

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

Signature: _____

Name: DU YULIN

Date: _____ / _____ / 2026